

An OpenAI model has disproved a central conjecture in discrete geometry

By: Hacker News

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What's New

****Claim****

An OpenAI model has disproved a central conjecture in discrete geometry.

****Evidence****

The claim is supported by the demonstration that the model generated solutions that contradict the conjecture, which had remained unproven for decades. Specific examples of these generated solutions were provided, indicating the model's ability to explore a vast solution space.

****Method****

The work utilized a large-scale generative model trained on discrete geometric configurations, employing a combination of heuristics and sampling techniques to explore potential counterexamples to the conjecture.

****Limitations****

The findings may be limited by the model's training data and inherent biases, raising questions about the generalizability of the results. There is also a potential reproducibility gap, as the specific conditions under which the model produced these counterexamples were not fully detailed.

****Safety relevance****

This development raises concerns about the reliability of AI-generated mathematical proofs and whether AI systems can be trusted to provide valid solutions in critical domains.

Full Announcement Details

An OpenAI model has disproved a central conjecture in discrete geometry. The claim is supported by the demonstration that the model generated solutions that contradict the conjecture, which had remained unproven for decades. Specific examples of these generated solutions were provided, indicating the model's ability to explore a vast solution space.

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<https://openai.com/index/model-disproves-discrete-geometry-conjecture/>

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